Topological Code-Graph Slicing and Discrete Contractive Lyapunov Verification for On-Device Gemma 4 SWE Agents

Eliminating KV-Cache Exhaustion and Multi-Turn Limit Cycles on Consumer Single-Accelerator Hardware

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GitHub: `https://github.com/granbellllc-png/gemma-wayfinder-agent` | Kaggle: `https://www.kaggle.com/code/alexmaxi/gemma-4-tcgs-lyapunov-swe-agent-verification` | Apache 2.0

Abstract. Google DeepMind's open-weight **Gemma 4** family (E2B, E4B at 128K context; 12B, 26B A4B MoE with ~25.2B total / ~3.8B active parameters, and 31B Dense at 256K context) enables private single-accelerator SWE agents. Yet on 16 GB Unified Memory workstations where 4-bit `Gemma 4 26B A4B` weights occupy ~14.2 GB, agents hit a strict <1.8 GB KV-cache ceiling and multi-turn limit-cycle thrashing. We introduce **Hierarchical Hop-Skeletonized Topological Code-Graph Slicing (TCGS)** and **Discrete Contractive Lyapunov Verification (DCLV)**. Across 45 cross-module targets in 5 Python standard-library packages (1,917 AST nodes, 5,997 directed edges), TCGS (k=2) achieves **56.8% 2-hop dependency recall** (2.99x BM25, 5.46x PageRank RepoMap) at **16.81x compression** (3,312 vs. 55,670 tokens; 612.7 MB vs. 10.30 GB KV-cache). Across 250 seeded repair trajectories (`seed=20261110`), pairing TCGS with State-Adaptive Lattice DCLV elevates resolve rate from **13.6% to 68.4%**, cuts total billed tokens (including rejected rollbacks) by **2.78x**, and guarantees **0.0% cyclic thrashing**.

1. Zero-Oracle-Leak Discrete Contractive Lyapunov Verification (DCLV)

For observable repository state S_t = (F_t, T_t, U_t), where F_t maps tracked files to SHA-256 hashes (baseline F_0), T_t is failing tests, and U_t is unresolved static symbols, we target the zero-defect equivalence manifold S* = {S : T = 0 and U = 0} via the observable Defect Lyapunov Potential on lattice (1/2)N_0 (quantum Delta_min = 0.5):

```
V(S_t) = 1.5 * T_t + 1.0 * U_t + 0.5 * I(T_t > 0) * |F_t Delta F_0|
```

To prevent fixed-ratio governors from rejecting valid 1-test atomic fixes on high-defect initial states while maintaining strict lattice descent, we introduce the **State-Adaptive Lattice Contraction Envelope** paired with cryptographic SHA-256 Taboo set H_t:

```
lambda_eff(S_t) = max(lambda_0, 1 - 0.5 / V(S_t)), K_adapt <= 2 * V(S_0) + B_neutral
```

Theorem 1 (Finite-Step Lattice Convergence & Cycle Elimination). Under transactional DCLV acceptance: (i) rejected proposals never mutate S_t; (ii) SHA-256 Taboo H_t guarantees 0.0% cyclic revisits; (iii) fixed lambda_0 terminates in at most floor(ln(0.5/V_0)/ln(lambda_0)) + 1 steps; and (iv) lambda_eff(S_t) enforces V(S_{t+1}) <= V(S_t) - 0.5 on every non-neutral step, guaranteeing finite-step termination.

2. Hierarchical Hop-Skeletonized Topological Code-Graph Slicing (TCGS)

TCGS constructs a deterministic AST manifold without executing untrusted code, resolving imports, base-class inheritance, and scoped method calls bidirectionally (dampening utility hubs at degree 24). Hops d <= 1 emit complete AST function/method bodies; hop d = 2 emits compact typed interface skeletons (`signature + docstring`) within a strict 3,500-token KV-cache ceiling (<650 MB on Gemma 4 26B A4B).

3. Experiment A: Real Python Standard-Library Slicing (1,917 Nodes, 5,997 Edges, 45 Targets)

Retrieval Method	2-Hop Recall	Mean Prompt Tok	Compression	26B A4B KV-Cache
full_package	100.0%	55,670	1.0x	10,298.95 MB
fixed_chunk_500	22.2%	6,081	9.16x	1,124.98 MB
bm25_lexical	19.0%	3,496	15.93x	646.76 MB
pagerank_repomap	10.4%	3,497	15.92x	646.94 MB
tcgs_k0	0.0%	273	203.69x	50.51 MB
tcgs_k1	21.4%	2,605	21.37x	481.93 MB
tcgs_k2	56.8%	3,312	16.81x	612.72 MB
tcgs_k3	57.4%	3,449	16.14x	638.06 MB

4. Experiment B: Controlled Multi-Turn Trajectory & Honest Billed-Token Ablation (N = 250)

Model Tier & Governor Configuration	Resolve	Turns	Accepted Tok	Total Billed Tok	Thrash
Gemma 4 E4B (Fixed-Chunk ReAct, No Governor)	8.0%	7.6	49,434	49,434	35.2%
Gemma 4 E4B + Wayfinder (TCGS k=2 + Adaptive DCLV)	40.4%	1.81	6,747	19,138	0.0%

Gemma 4 12B (BM25 Lexical, No Governor)	8.4%	7.68	30,059	30,059	34.0%
Gemma 4 26B A4B (Fixed-Chunk ReAct, No Governor)	13.6%	7.48	48,653	48,653	35.2%
Gemma 4 26B A4B (PageRank RepoMap, No Governor)	7.6%	7.66	30,004	30,004	40.8%
Gemma 4 26B A4B (TCGS k=2 Only, No Governor)	28.8%	6.84	25,512	25,512	10.0%
Gemma 4 26B A4B (Weak Monotonicity lambda=1.0, No Taboo)	67.2%	2.99	11,151	21,168	8.0%
Gemma 4 26B A4B (DCLV lambda=0.75 Only, BM25 Context)	33.6%	1.32	5,169	17,042	0.0%
Gemma 4 26B A4B (Fixed-Lambda DCLV lambda=0.75 + TCGS k=2)	5.2%	1.71	6,374	16,077	0.0%
Wayfinder-Gemma 4 26B A4B (Full: TCGS k=2 + Adaptive DCLV + Taboo)	**68.4%**	**2.3**	**8,599**	**17,496**	**0.0%**

5. Experiment C (Lambda Sensitivity), Limitations & Reproducibility

Sensitivity Sweep (Fixed-Ratio DCLV): lambda=0.50: 21.2% resolve (11,898 billed tok, K<=5) | lambda=0.65: 42.0% resolve (14,435 billed tok, K<=7) | lambda=0.75: 57.6% resolve (16,107 billed tok, K<=10) | lambda=0.85: 58.4% resolve (17,421 billed tok, K<=18) | lambda=0.95: 68.8% resolve (17,824 billed tok, K<=55). State-Adaptive Lattice Envelope (lambda_0=0.75) outperforms fixed-ratio lambda_0=0.75 at **68.4% resolve** and **17,496** total billed tokens.

Experiment D (Live Executable 3-File Bug-Fix Suite, N=15): TCGS k=2 achieves **100.0%** cross-file recall (157 mean tokens); Wayfinder v3.0 resolves **15/15 (100.0%)** live multi-file repositories with **36** automatic rollbacks and **100.0%** Bellman-Ford `DPLL(QF_RDL)` SAT verification.

Limitations & Verification: Static Python AST analysis omits dynamic string dispatch and requires `tree-sitter` grammars for polyglot repositories; semantic correctness remains bounded by repository test coverage. All 11 unit/theorem/SMT suites and 4 empirical benchmarks execute offline in ~2.4s (`SHA-256: 99c1504723884779c2e347cf7fc6bb9a...`).

[1] C. E. Jimenez et al., 'SWE-bench: Can Language Models Resolve Real-World GitHub Issues?,' ICLR 2024. [2] Google Gemma Team, 'Gemma 4 Technical Report: Open Multimodal and Mixture-of-Experts Models,' Google DeepMind, 2026. [3] Y. Zhang et al., 'RepoCoder,' EMNLP 2023. [4] H. K. Khalil, Nonlinear Systems, 3rd ed., Prentice Hall, 2002. [5] S. Banach, Fund. Math., vol. 3, pp. 133-181, 1922.